

where

$$\begin{aligned} M_{11} &= \frac{1}{3}m_1a_1^2 + m_2 \left(a_1^2 + a_1a_2c_2 + \frac{1}{3}a_2^2 \right) + m_3 (a_1^2 + 2a_1a_2c_2 + a_2^2) \\ M_{12} &= m_2 \left(\frac{1}{2}a_1a_2c_2 + \frac{1}{3}a_2^2 \right) + m_3 (a_1a_2c_2 + a_2^2) \\ M_{22} &= \frac{1}{3}m_2a_2^2 + m_3a_2^2 \end{aligned}$$

The gravity vector $\mathbf{g}(\mathbf{q})$ can also be easily calculated based on the Jacobian matrices of the links as follows.

$$\mathbf{g}(\mathbf{q}) = - \sum_{i=1}^3 m_i \mathbf{J}_{v_i}^T \mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -m_3g \end{bmatrix}$$

The vector of Coriolis and centrifugal forces $\mathbf{v}(\mathbf{q}, \dot{\mathbf{q}})$ is also equal to:

$$\mathbf{v}(\mathbf{q}, \dot{\mathbf{q}}) = \left(\dot{\mathbf{M}}(\mathbf{q})\dot{\mathbf{q}} - \frac{\partial K}{\partial \mathbf{q}} \right) = \left(\frac{1}{2}m_2 + m_3 \right) a_1a_2s_2 \begin{bmatrix} \dot{\theta}_2^2 + 2\dot{\theta}_1\dot{\theta}_2 \\ \dot{\theta}_1^2 \\ 0 \end{bmatrix}$$

By determining the necessary matrices and vectors, the dynamic motion equations of the robot can be represented in the following closed-form.

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{v}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) = \mathbf{Q}$$

8.5.4 3DOF Elbow Manipulator

Case Study 8.5 (3DOF Elbow Manipulator Robot) Consider the 3DOF Elbow manipulator shown in [Figure 8.9](#). Suppose the first link of the robot has a mass of m_1 and an inertia moment $I_{zz} = I_1$. Consider the second and third links of the robot as slender rods. Assume the center of mass for all links is at their midpoint and neglect friction in the robot joints. Utilize the recursive Euler-Lagrange method to determine the dynamic formulation of the robot and calculate its mass matrix, gravity, and Coriolis and centrifugal vectors.

Solution: Considering the information provided, the generalized coordinates for this robot can be represented as $\mathbf{q} = [\theta_1, \theta_2, \theta_3]^T$. The generalized force vector, based on the generalized coordinates, is denoted as $\mathbf{Q} = [\tau_1, \tau_2, \tau_3]^T$. Assuming slender rods for the links, the inertia matrix for these links with respect to their center of mass can be obtained.

$$\begin{aligned} {}^C\mathbf{I}_1 &= I_1 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^C\mathbf{I}_2 = \frac{1}{12}m_2a_2^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ {}^C\mathbf{I}_3 &= \frac{1}{12}m_3a_3^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Now, using the rotation matrix, the inertia moments of the links in the reference coordinate system 0 are obtained as follows:

$$\begin{aligned}
 {}^0\mathbf{I}_1 &= {}^0\mathbf{R}_1 {}^C\mathbf{I}_1 {}^0\mathbf{R}_1^T = I_1 \begin{bmatrix} s_1^2 & -s_1 c_1 & 0 \\ -s_1 c_2 & c_1^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 {}^0\mathbf{I}_2 &= {}^0\mathbf{R}_2 {}^C\mathbf{I}_2 {}^0\mathbf{R}_2^T = \frac{1}{12} m_2 a_2^2 \begin{bmatrix} c_1^2 s_2^2 + s_1^2 & -c_1 s_1 c_2^2 & -c_1 c_2 s_2 \\ -c_1 s_1 c_2^2 & s_1^2 s_2^2 + c_1^2 & -s_1 c_2 s_2 \\ -c_1 c_2 s_2 & -s_1 c_2 s_2 & c_2^2 \end{bmatrix} \\
 {}^0\mathbf{I}_3 &= {}^0\mathbf{R}_3 {}^C\mathbf{I}_3 {}^0\mathbf{R}_3^T = \frac{1}{12} m_3 a_3^2 \begin{bmatrix} c_1^2 s_{23}^2 + s_1^2 & -c_1 s_1 c_{23}^2 & -c_1 c_{23} s_{23} \\ -c_1 s_1 c_{23}^2 & s_1^2 s_{23}^2 + c_1^2 & -s_1 c_{23} s_{23} \\ -c_1 c_{23} s_{23} & -s_1 c_{23} s_{23} & c_{23}^2 \end{bmatrix}
 \end{aligned}$$

Next, it is necessary to calculate the Jacobian matrices of the links. For this purpose, the vectors representing the distance from the center of mass of the links to their corresponding joints with respect to the reference coordinate system $\{0\}$ are determined using recursive calculations.

$${}^0\mathbf{p}_{c_1}^0 = \begin{bmatrix} 0 \\ 0 \\ -d_1/2 \end{bmatrix}, {}^1\mathbf{p}_{c_2}^1 = {}^0\mathbf{p}_{c_2}^0 = \frac{a_2}{2} \begin{bmatrix} c_1 c_2 \\ s_1 c_2 \\ s_2 \end{bmatrix}$$

Similarly,

$${}^2\mathbf{p}_{c_3}^2 = \frac{a_3}{2} \begin{bmatrix} c_1 c_{23} \\ s_1 c_{23} \\ s_{23} \end{bmatrix} \rightarrow {}^1\mathbf{p}_{c_3}^1 = {}^0\mathbf{p}_{c_3}^0 = \begin{bmatrix} c_1 \left(a_2 c_2 + \frac{a_3}{2} c_{23} \right) \\ s_1 \left(a_2 c_2 + \frac{a_3}{2} c_{23} \right) \\ a_2 s_2 + \frac{a_3}{2} s_{23} \end{bmatrix}$$

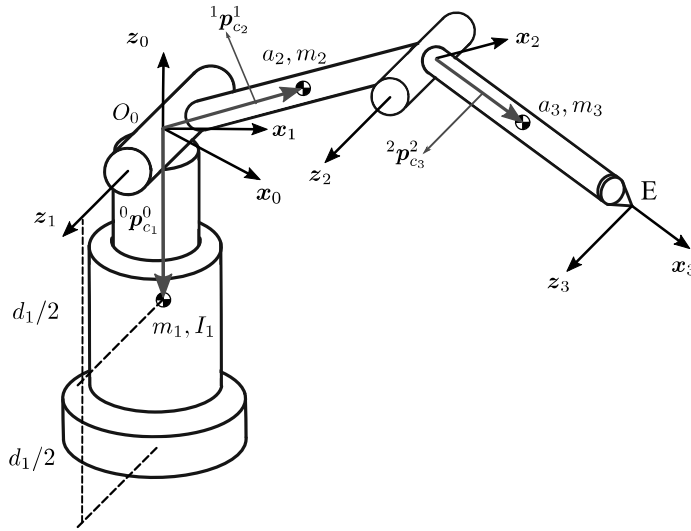


FIGURE 8.9

Schematic of a 3DOF Elbow manipulator and the representation of necessary vectors in determining the Jacobian matrix of its links.

By this means, the Jacobian matrices of the links can be easily obtained.

$$\begin{aligned} \mathbf{J}_{v_1} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{J}_{v_2} = \begin{bmatrix} -\frac{1}{2}a_2s_1c_2 & -\frac{1}{2}a_2c_1s_2 & 0 \\ \frac{1}{2}a_2c_1c_2 & -\frac{1}{2}a_2s_1s_2 & 0 \\ 0 & \frac{1}{2}a_2c_2 & 0 \end{bmatrix} \\ \mathbf{J}_{v_3} &= \begin{bmatrix} -s_1(a_2c_2 + \frac{1}{2}a_3c_{23}) & -c_1(a_2s_2 + \frac{1}{2}a_3s_{23}) & -\frac{1}{2}a_3c_1s_{23} \\ c_1(a_2c_2 + \frac{1}{2}a_3c_{23}) & -s_1(a_2s_2 + \frac{1}{2}a_3s_{23}) & -\frac{1}{2}a_3s_1s_{23} \\ 0 & a_2c_2 + \frac{1}{2}a_3c_{23} & \frac{1}{2}a_3c_{23} \end{bmatrix} \\ \mathbf{J}_{\omega_1} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{J}_{\omega_2} = \begin{bmatrix} 0 & s_1 & 0 \\ 0 & -c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathbf{J}_{\omega_3} = \begin{bmatrix} 0 & s_1 & s_1 \\ 0 & -c_1 & -c_1 \\ 1 & 0 & 0 \end{bmatrix} \end{aligned}$$

Now, by using the Jacobian matrices of the links, the symmetric mass matrix of the robot can be determined.

$$\mathbf{M}(\mathbf{q}) = \sum_{i=1}^3 m_i \left(\mathbf{J}_{v_i}^T \mathbf{J}_{v_i} + \mathbf{J}_{\omega_i}^T \mathbf{I}_i \mathbf{J}_{\omega_i} \right) = \begin{bmatrix} M_{11} & 0 & 0 \\ & M_{22} & M_{23} \\ & & M_{33} \end{bmatrix}$$

where

$$\begin{aligned} M_{11} &= \left(\frac{1}{3}m_2 + m_3 \right) a_2^2 c_2^2 + \frac{1}{3}m_3 a_3^2 c_{23}^2 + m_3 a_2 a_3 c_2 c_{23} \\ M_{22} &= \left(\frac{1}{3}m_2 + m_3 \right) a_2^2 + \frac{1}{3}m_3 a_3^2 + m_3 a_2 a_3 c_3 \\ M_{23} &= \frac{1}{3}m_3 a_3^2 + \frac{1}{2}m_3 a_2 a_3 c_3 \\ M_{33} &= \frac{1}{3}m_3 a_3^2 \end{aligned}$$

The gravity vector $\mathbf{g}(\mathbf{q})$ can also be calculated based on the Jacobian matrices of the links as follows.

$$\mathbf{g}(\mathbf{q}) = - \sum_{i=1}^3 m_i \mathbf{J}_{v_i}^T \mathbf{g} = \begin{bmatrix} \frac{1}{2}m_2 a_2 c_1 c_2 + m_3 c_1 (a_2 c_2 + \frac{1}{2}a_3 c_{23}) \\ -\frac{1}{2}m_2 a_2 s_1 s_2 - m_3 s_1 (a_2 s_2 + \frac{1}{2}a_3 s_{23}) \\ -\frac{1}{2}m_3 a_3 s_1 s_{23} \end{bmatrix} \mathbf{g}.$$

The vector of Coriolis and centrifugal forces $\mathbf{v}(\mathbf{q}, \dot{\mathbf{q}})$ is also determined through symbolic calculations.

$$\mathbf{v}(\mathbf{q}, \dot{\mathbf{q}}) = \left(\dot{\mathbf{M}}(\mathbf{q}) \dot{\mathbf{q}} - \frac{\partial K}{\partial \mathbf{q}} \right) = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \mathbf{v}_2 \end{bmatrix}$$

where

$$\begin{aligned} \mathbf{v}_1 &= -\frac{1}{3}m_2 a_2^2 \sin(2\theta_2) \dot{\theta}_1 \dot{\theta}_2 \\ &\quad - m_3 \left(\frac{1}{3}a_3^2 \sin 2(\theta_2 + \theta_3) + a_2 a_3 \sin(2\theta_2 + \theta_3) + a_2^2 \sin 2\theta_2 \right) \dot{\theta}_1 \dot{\theta}_2 \\ &\quad - m_3 \left(\frac{1}{3}a_3^2 \sin 2(\theta_2 + \theta_3) + \frac{1}{2}a_2 a_3 \sin(2\theta_2 + \theta_3) + \frac{1}{2}a_2 a_3 s_3 \right) \dot{\theta}_1 \dot{\theta}_3 \end{aligned}$$

$$\begin{aligned}
\mathbf{v}_2 &= m_3 \left(\frac{1}{3} a_3^2 c_{23} s_{23} + \frac{1}{2} a_2 a_3 c_{23} s_2 + \frac{1}{2} a_2 a_3 s_{23} c_2 \right) \dot{\theta}_1^2 \\
&\quad + \left(\frac{1}{3} m_2 + m_3 \right) a_2^2 s_2 c_2 \dot{\theta}_1^2 - m_3 a_2 a_3 s_3 \dot{\theta}_2 \dot{\theta}_3 - \frac{1}{2} m_3 a_2 a_3 s_3 \dot{\theta}_3^2 \\
\mathbf{v}_3 &= \frac{1}{2} m_3 a_2 a_3 s_{23} c_2 \dot{\theta}_1^2 + \frac{1}{2} m_3 a_2 a_3 s_3 \dot{\theta}_2^2 + \frac{1}{6} m_3 a_3^2 \sin 2(\theta_2 + \theta_3) \dot{\theta}_1^2
\end{aligned}$$

By determining the necessary matrices and vectors, the dynamic formulation of this robot can be written in the following closed-form.

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{v}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) = \mathbf{Q}$$

Problems

- 8.1 Consider the two-degrees-of-freedom RP robot shown in [Figure 8.10](#). Assume the robot links as slender rods with their center of mass located at their midpoint lengths, and consider viscous friction in the joints. Using the Euler-Lagrange method, derive the dynamic equations of the robot in closed-form and obtain the robot's mass matrix $\mathbf{M}(\mathbf{q})$, gravitational vector $\mathbf{g}(\mathbf{q})$, and the vector for Coriolis and centrifugal torques $\mathbf{v}(\mathbf{q}, \dot{\mathbf{q}})$.

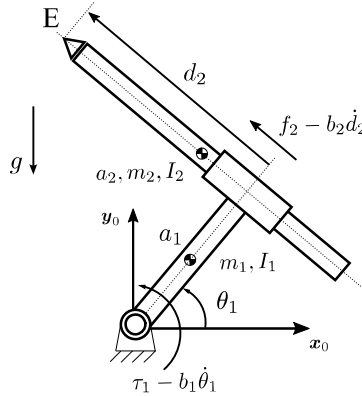


FIGURE 8.10
2-DOF robot of Problem 8.1

- 8.2 Consider the three-degrees-of-freedom PRP robot illustrated in [Figure 8.11](#). Assume the robot links are slender rods with their center of mass located at their midpoints, and consider viscous friction in the joints. Utilizing the Euler-Lagrange method, deduce the dynamic equations of the robot in closed-form and determine the robot's mass matrix $\mathbf{M}(\mathbf{q})$, gravitational vector $\mathbf{g}(\mathbf{q})$, and the vector representing Coriolis and centrifugal torques $\mathbf{v}(\mathbf{q}, \dot{\mathbf{q}})$.
- 8.3 Consider the 3-DOF PRR robot depicted in [Figure 8.12](#). Assume the robot links are slender rods with their center of mass located at the midpoint of their lengths, and consider viscous friction in the joints. Using the Euler-Lagrange method, derive the